TalentScout AI - Intelligent Hiring Assistant

PG-AGI

ASSIGNMENT FOR AI/ML INTERN

BY

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1. INTRODUCTION

TalentScout AI is an intelligent, AI-powered hiring assistant designed to streamline and automate the initial screening process for technical candidates. Traditional recruitment workflows often involve repetitive manual screening, inconsistent evaluations, and high time investment. This project addresses these inefficiencies by leveraging Large Language Models (LLMs), conversational AI, and sentiment analysis to create a smart, scalable, and interactive hiring solution.

The system simulates a recruiter by conducting structured, multi-turn conversations with candidates, collecting essential information, and dynamically generating technical interview questions tailored to the candidate's expertise. By integrating automation with intelligent decision-making, TalentScout AI enhances both recruiter productivity and candidate experience.

2. OBJECTIVES

The primary objectives of this project are -

- Automate the initial screening of candidates for technical roles.
- Provide a conversational interface that mimics real human interaction.
- Generate dynamic, role-specific technical questions based on candidate input.
- Analyze candidate responses using sentiment analysis.
- Store structured candidate data for further evaluation and decision-making.

3. SYSTEM ARCHITECTURE

The system follows a modular and scalable architecture -

User → Streamlit Interface → Session State Manager → Prompt Engine → LLM (Groq) → Response Generator → UI Display → Database Storage

Key Components -

- Frontend (Streamlit) - Provides an interactive chatbot interface.
- Session State Manager - Maintains conversation history and context.
- Prompt Engine - Constructs structured prompts for different stages.
- LLM (Groq - LLaMA 3.1) - Generates intelligent responses and questions.
- Sentiment Analyzer (VADER) - Evaluates user responses.
- Database (SQLite) - Stores candidate data.
- JSON Export - Enables data portability.

4. KEY FEATURES

4.1 AI-Powered Conversational Interface

- Multi-turn conversation with context awareness.
- Step-by-step candidate data collection.
- Natural and engaging interaction flow.

4.2 Dynamic Technical Question Generation

- Generates 3–5 questions per technology.
- Adapts difficulty based on experience level.
- Includes scenario-based and real-world questions.

4.3 Context-Aware Interaction

- Maintains full chat history using session state.
- Ensures continuity and relevance in responses.

4.4 Sentiment Analysis

- Uses VADER to classify responses -
  - Positive.
  - Neutral.
  - Negative.
- Helps assess candidate confidence and clarity.

4.5 Tech Stack Normalization

- Converts informal inputs into standardized forms -
  - "py" → Python
  - "reactjs" → React
- Ensures consistency in question generation.

4.6 Data Persistence

- Stores candidate details in SQLite database.
- Exports structured data in JSON format.

4.7 Enhanced UI/UX

- Clean chat interface.
- Progress tracking bar.
- Loading indicators for AI responses.
- Sidebar for feature overview.

5. PROMPT ENGINEERING STRATEGY

The system uses multiple specialized prompts to ensure structured and effective interaction -

5.1 Information Gathering Prompt

Guides the chatbot to collect -

- Name.
- Email.
- Phone number.
- Experience.
- Desired role.
- Location.
- Tech stack.

5.2 Question Generation Prompt

- Generates tailored technical questions.
- Adapts complexity based on experience level.
- Focuses on practical and scenario-based problems.

5.3 Contextual Prompt

- Uses full conversation history.
- Maintains coherence across interactions.

5.4 Fallback Prompt

- Handles unexpected or irrelevant inputs.
- Redirects conversation back to the hiring flow.

6. WORKFLOW

1. User initiates conversation via chatbot.
2. System collects candidate details step-by-step.
3. Candidate provides their tech stack.
4. System -
    - Normalizes technologies.
    - Generates relevant technical questions.
5. Candidate responds to questions.
6. System performs sentiment analysis on responses.
7. All data is stored in SQLite and exported as JSON.
8. Conversation concludes with a summary.

7. TECHNOLOGY STACK

Component	Technology Used
Frontend	Streamlit
Backend	Python
LLM	Groq (LLaMA 3.1 Models)
NLP	VADER Sentiment Analysis
Database	SQLite
Deployment	ngrok
Data Handling	JSON

8. CHALLENGES AND SOLUTIONS

API Cost Constraints

- Solution - Used Groq API with free-tier LLaMA models.

Context Management

- Solution - Maintained session-based conversation history.

Model Deprecation

- Solution - Implemented dynamic model selection and fallback mechanisms.

Environment Variable Handling

- Solution -Added flexible API key handling for local and Colab environments.

9. FUTURE ENHANCEMENTS

- Resume upload and parsing using RAG pipelines.
- Recruiter admin dashboard.
- Multi-language conversational support.
- Voice-based interaction system.
- Cloud deployment (AWS/GCP).

10. CONCLUSION

TalentScout AI demonstrates the practical application of AI and machine learning in transforming recruitment processes. By combining conversational AI, LLMs, and intelligent automation, the system reduces manual effort, improves candidate engagement, and ensures consistent evaluation.

This project highlights how modern AI systems can be integrated into real-world business workflows, making hiring faster, smarter, and more efficient.